Research proposal

Individual proposal template in four parts, built in Sessions 1–7 and ending with a publication plan

Foundations of Clinical Research · Template version 1.0

Proposal title: *[Working title (see 1.1)]*
Author: *[Your name (profession)]*
Department: *[Department]*, *[Institution]*
Version / date: *[Version 1]* · *[DD/MM/YYYY]*

How to use this template

- Replace every [grey italic text in brackets] with your own text. Press Ctrl + H and search for "[" to step through them, and check none remain before you share the document.
- Grey italic guidance and shaded "Example" boxes are there to help you. Delete them as you write: Home → Styles → right-click "Guidance" (or "Example") → Select All → Delete.
- Each part is modelled live by the instructor in the proposal-builder segment of a session, and you write your own as homework: Part 1 (Sessions 1–2), Part 2 (Sessions 3–4), Part 3 (Sessions 5–6), Part 4 (Session 7). Everyone submits it in Session 8; about 5 volunteers also present it. Keep the finished proposal to about 6–8 pages.
- The shaded examples use the course running example: *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?*
- Choose your own question from your department. Use the examples to see what a good answer looks like, not as text to copy.

Part 1. Ask: the research question (Sessions 1–2)

1.1 Working title

Build it from your PICO: 15–20 words saying who (population), what (intervention or exposure), which outcome and, ideally, the design. Avoid abbreviations and questions. It is a WORKING title: you will refine it in the publication plan (4.5) and it will become the title of your paper.

Example: *Restrictive versus usual fluid strategy and 28-day mortality in adults with septic shock: a pilot randomised trial in the ICU of [Institution].*

[Your title]

1.2 Problem statement and rationale

150–250 words in four steps: (1) the problem and how big it is (numbers); (2) what is already known; (3) what is NOT known (the gap); (4) why answering it matters for your patients or unit. Cite your key references by number [1].

Example: *Septic shock is common in our ICU and many patients die within a month. Patients receive large volumes of fluid, and a positive fluid balance has been linked to higher mortality in cohort studies [2]. A small trial showed that limiting fluids is possible [1], but whether it saves lives is still unknown. Practice in our unit varies widely between doctors ...*

[Problem, what is known, the gap, why it matters]

1.3 PICO

Break your question into its parts. For observational studies, "I" is the exposure. T (time) is the time point at which the outcome is measured.

	Element	Our study	Example (running example)
P	Population / patients	*[Who?]*	*Adults (≥18 y) with septic shock in the ICU*
I	Intervention / exposure	*[What is done or what are they exposed to?]*	*Restrictive fluid strategy*
C	Comparison	*[Compared with what?]*	*Usual care*
O	Outcome	*[What will you measure?]*	*All-cause mortality*
T	Time	*[When?]*	*At 28 days*

1.4 Research question

One sentence, ending with a question mark, that contains all PICO(T) elements.

> **Example:** *In adults with septic shock in the ICU, does a restrictive fluid strategy, compared with usual care, reduce 28-day mortality?*

[Our research question]

1.5 FINER check

Is the question worth answering and can you and your department answer it? Be honest: a "No" is a reason to narrow or change the question, not a failure.

	Criterion	Question to ask	Yes / Partly / No	Comment
F	Feasible	Enough patients, time, skills, money and data?	☐ Y ☐ P ☐ N	*[Why?]*
I	Interesting	Do clinicians in your unit want the answer?	☐ Y ☐ P ☐ N	*[Why?]*
N	Novel	Does it confirm, refute or extend what is known (see 1.2)?	☐ Y ☐ P ☐ N	*[Why?]*
E	Ethical	Can it be done without undue risk to patients?	☐ Y ☐ P ☐ N	*[Why?]*
R	Relevant	Would the answer change practice, policy or future research?	☐ Y ☐ P ☐ N	*[Why?]*

> **Example:** *Feasible: partly. Our ICU admits about 15 patients with septic shock a month. A mortality trial would need several hundred patients, so we propose a pilot.*

1.6 Aim, objectives and hypothesis

Aim = the overall purpose (one sentence). Objectives = 2–3 specific, measurable steps (start with a verb: to compare, to estimate, to describe). Hypothesis = what you expect, written so that it can be tested.

Aim: *[To ...]*

Primary objective: *[To compare / estimate ...]*

Secondary objectives: *[1. To ... 2. To ...]*

Hypothesis (H_1): *[We expect that ...]*

Null hypothesis (H_0): *[There is no difference in ...]*

> **Example:** *Primary objective: to compare 28-day mortality between patients managed with a restrictive fluid strategy and those receiving usual care. H_0 : 28-day mortality does not differ between the two strategies.*

1.7 Five key references

The five most relevant papers from your Literature Review Worksheet, in Vancouver style (numbered in the order they are cited).

> **Example:** *Boulet N, Quenot J-P, Serrand C, et al. Impact on fluid balance of an optimized restrictive strategy targeting non-resuscitative fluids in intensive care patients with septic shock: a single-blind, multicenter, randomized, controlled, pilot study. Crit Care. 2024;28:429. doi:10.1186/s13054-024-05155-z*

1. *[Reference 1: Authors. Title. Journal. Year;Volume:Pages. doi]*
2. *[Reference 2: Authors. Title. Journal. Year;Volume:Pages. doi]*
3. *[Reference 3: Authors. Title. Journal. Year;Volume:Pages. doi]*
4. *[Reference 4: Authors. Title. Journal. Year;Volume:Pages. doi]*
5. *[Reference 5: Authors. Title. Journal. Year;Volume:Pages. doi]*

Part 2. Design: how the study will be done (Sessions 3–4)

2.1 Study design and justification

Tick one design, then explain in 2–3 sentences why it is the best feasible design for your question (use the design decision tree from Session 3).

☐ Cross-sectional ☐ Case-control ☐ Prospective cohort ☐ Retrospective cohort ☐ Randomised controlled trial ☐ Other: *[specify]*

> **Example:** *A pilot randomised controlled trial, because randomisation balances illness severity between groups (sicker patients receive more fluid: confounding by indication) and a pilot tells us whether a full trial is feasible.*

[Our design and why]

2.2 Setting

Where (unit, number of beds, type of hospital) and when (study period).

[e.g. The [number]-bed adult ICU of [Institution], from [month/year] to [month/year]]

2.3 Study population

Inclusion criteria define who CAN enter; exclusion criteria remove people who meet the inclusion criteria but should not take part (safety, impossible to measure the outcome...).

Inclusion criteria	Exclusion criteria
Age ≥ 18 years	*Renal replacement therapy already started*
Septic shock (Sepsis-3 definition) within 24 h of starting vasopressors	*Pregnancy; expected death within 24 h*
[Inclusion 1]	*[Exclusion 1]*
[Inclusion 2]	*[Exclusion 2]*
[Inclusion 3]	*[Exclusion 3]*

2.4 Intervention or exposure, and comparator

Describe exactly what the intervention (trial) or exposure (observational study) is, and what it is compared with, so that another unit could do the same.

Intervention / exposure: *[What, how much, how often, who gives it, for how long]*

Comparator: *[e.g. usual care as decided by the treating team]*

2.5 Outcomes

ONE primary outcome (it decides the sample size) and a few secondary outcomes. For each: how it is measured and when.

Type	Outcome	How measured / definition	When
Primary	*All-cause mortality*	*Alive / dead, from medical records or phone call*	*Day 28*
Primary	*[Outcome]*	*[Definition]*	*[Time point]*
Secondary	*[Outcome]*	*[Definition]*	*[Time point]*
Secondary	*[Outcome]*	*[Definition]*	*[Time point]*

2.6 Main threats of bias and how they are reduced

Name 2–3 biases that could give a wrong answer in YOUR study and what you will do about each one (randomisation, blinding, matching, adjustment, standard definitions, training ...).

Bias	How it could affect our result	What we will do
Confounding by indication	*Sicker patients receive more fluid AND die more*	*Randomise; record SOFA and lactate to check balance*
Observer (information) bias	*Staff who know the group may record fluids differently*	*Standard fluid chart; outcome (death) is objective*
[Bias 1]	*[Effect]*	*[Action]*
[Bias 2]	*[Effect]*	*[Action]*
[Bias 3]	*[Effect]*	*[Action]*

Part 3. Measure: data and analysis (Sessions 5–6)

3.1 Variables table

List every variable you need: the outcome, the exposure/intervention, the main confounders and a few descriptive variables. Type: continuous, binary, categorical (nominal or ordinal), date.

Variable	Role	Type	Unit or categories	Source
28-day mortality	*Outcome*	*Binary*	*0 = alive, 1 = dead*	*Medical record / phone*
Cumulative fluid balance	*Exposure*	*Continuous*	*ml/kg*	*Fluid chart*
SOFA score	*Confounder*	*Ordinal / discrete*	*0–24 points*	*ICU chart*
Lactate at admission	*Confounder*	*Continuous*	*mmol/L*	*Laboratory*
[Variable]	*[Role]*	*[Type]*	*[Unit]*	*[Source]*
[Variable]	*[Role]*	*[Type]*	*[Unit]*	*[Source]*
[Variable]	*[Role]*	*[Type]*	*[Unit]*	*[Source]*
[Variable]	*[Role]*	*[Type]*	*[Unit]*	*[Source]*

3.2 Data collection tool: mini-CRF (6–10 items)

Write each question exactly as it will appear on the case report form. One item per question, units stated, coded tick boxes, a "not done / unknown" option, dates as DD/MM/YYYY, and a participant ID instead of a name. See CRF_Template.docx.

#	Question (CRF item)	Answer format and codes
1	*Participant ID*	|__|__|-|__|__|__| *(site, then number)*
2	*Age at ICU admission*	|__|__|__| *years*
3	*Cumulative fluid balance, day 1–5*	|__|__|__|__|__| *ml (+ or −)* ☐ *not done*
4	*Vital status at day 28*	☐ *0 Alive* ☐ *1 Dead* ☐ *9 Unknown*
5	*[Item]*	*[Format, unit, codes]*
6	*[Item]*	*[Format, unit, codes]*
7	*[Item]*	*[Format, unit, codes]*
8	*[Item]*	*[Format, unit, codes]*

3.3 Data dictionary (key rows)

One row per variable in your database. Short variable names without spaces; the same missing-value codes for every variable.

Variable name	Label	Type	Units / codes	Valid range	Missing code
cfb_kg_d5	*Cumulative fluid balance to day 5*	*Continuous*	*ml/kg*	*−150 to +300*	*−99*
vital_d28	*Vital status at day 28*	*Binary*	*0 = alive, 1 = dead*	*0, 1*	*−99*
[name]	*[label]*	*[type]*	*[units]*	*[range]*	*[code]*
[name]	*[label]*	*[type]*	*[units]*	*[range]*	*[code]*
[name]	*[label]*	*[type]*	*[units]*	*[range]*	*[code]*

3.4 Data management and quality plan

How will data travel from the patient to the analysis file, and how will you keep them complete and correct? Think: who collects; paper CRF or electronic data capture (e.g. REDCap, KoBoToolbox); training and a CRF pilot; range and consistency checks; double entry of a sample; audit trail; back-ups and version control; where data are stored and for how long; a data-sharing statement.

Step	Our plan
Collection and tool	*[e.g. trained ICU nurses, daily, paper CRF then electronic database]*
Entry and checks	*[e.g. range/consistency checks; 10% double entry; weekly query list]*
Storage, back-up, access	*[e.g. institution server, daily back-up, named users with passwords]*
Data sharing	*[e.g. de-identified data available on reasonable request after publication]*

> **Example:** *Two trained ICU nurses collect data daily; the CRF is piloted on 5 patients; the electronic database rejects values outside the valid range and keeps an audit trail; 10% of CRFs are entered twice and compared; the database is backed up daily on the [Institution] server.*

3.5 Sample size statement

Use the online calculator from the Session 6 demo. Fill in the sentence below. If the number is not feasible, say so and propose a pilot or a multicentre study.

To detect a difference in *[primary outcome]* from *[expected value in comparison group]* to *[expected value in intervention group]*, with 80% power and a 5% significance level (two-sided), we need *[n]* participants per group. Allowing for *[10]*% loss to follow-up, we will recruit *[N]* participants in total.

> **Example:** *To detect a fall in 28-day mortality from 40% to 30% we need 356 patients per group (712 in total; about 790 allowing 10% loss). This is not feasible in one ICU in one year, so we propose a pilot of 60 patients to test recruitment and adherence.*

3.6 Analysis plan in plain language

For each objective: which variable, which summary (mean (SD), median (IQR), n (%)), and which test or method (use the test decision table from Session 6).

Objective	Variable(s)	Summary	Test / method
Describe the patients (Table 1)	*Age, sex, SOFA, lactate*	*Mean (SD) or median (IQR); n (%)*	*None (descriptive)*
Compare 28-day mortality	*vital_d28 by group*	*n (%) per group*	*Chi-square test; risk ratio with 95% CI*
Compare fluid balance	*cfb_kg_d5 by group*	*Median (IQR)*	*Mann–Whitney U test*
[Objective]	*[Variables]*	*[Summary]*	*[Test]*
[Objective]	*[Variables]*	*[Summary]*	*[Test]*

3.7 Missing-data plan

Prevention first (Session 5.1), then a plan written BEFORE you see the data: how you will report missingness, why data might be missing (completely at random, at random, not at random), what the main analysis does (complete-case or multiple imputation) and which sensitivity analysis checks it. See Statistical_Analysis_Plan_Template.docx.

Prevention: *[e.g. mandatory fields, daily checks while the patient is in the ICU, phone follow-up]*

Reporting: *[e.g. number (%) missing for every variable, in Table 1 and the flow diagram]*

Main analysis: *[e.g. complete-case if < 5% missing; otherwise multiple imputation]*

Sensitivity analysis: *[e.g. best/worst case for missing vital status]*

Part 4. Protect and publish (Session 7)

4.1 Ethics and consent

Which ethics committee will review the study? What are the risks and benefits for participants? Who gives consent, and how? In the ICU many patients cannot consent: say whether you will ask a legal representative, use deferred consent, or request a waiver, and why.

Ethics committee: *[Ethics committee name]*

Risks / benefits: *[Main risks, how they are reduced; expected benefits]*

Consent approach: ☐ Written, by the patient ☐ Legal representative ☐ Deferred consent ☐ Waiver requested

Why this approach: *[Justification, referring to [National regulations]]*

4.2 Registration

Trials must be registered in a public registry BEFORE the first participant is enrolled, or most journals will refuse the paper (ICMJE). Registration is recommended for observational studies; systematic reviews are registered in PROSPERO.

Registry: ☐ ClinicalTrials.gov ☐ A WHO ICTRP primary registry ☐ PROSPERO (reviews) ☐ Other / not required

When / number: *[Planned date (before first participant)]* · *[Registration ID]*

4.3 Confidentiality

How will you protect participants' identity? Think: study ID instead of name, a separate locked key file, password-protected database, who has access, how long data are kept.

[Our confidentiality measures]

4.4 Feasibility

Timeline (Gantt chart)

Shade the months for each activity (select the cells → Table Design → Shading). Ethics approval usually takes longer than expected.

Activity	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Ethics approval												
CRF pilot and staff training												
Recruitment and data collection												
Data cleaning and analysis												
Writing and submission												
[Activity]												
[Activity]												

Budget headings

List the costs, even if the hospital covers them. Only include tests that are not part of routine care.

Budget heading	What it covers	Amount
Personnel	*[e.g. research nurse or data collector time]*	*[0]*
Training	*[e.g. GCP and CRF training]*	*[0]*
Materials and printing	*[e.g. CRFs, consent forms]*	*[0]*
Tests or equipment	*[only if not routine care]*	*[0]*
Data management and statistics	*[e.g. database, statistician time]*	*[0]*
Ethics and registration fees	*[as charged by the committee / registry]*	*[0]*
Dissemination	*[e.g. open-access fee, conference]*	*[0]*
Contingency (5–10%)	*[unexpected costs]*	*[0]*
Total (*[currency]***)**		*[0]*

Resources and recruitment

Question	Our answer
How many eligible patients per month? (source of the number)	*[e.g. 15/month, ICU register]*
How many will agree / be included?	*[e.g. about 70%]*
Who collects the data, and when?	*[Names / roles]*
Are the data routinely recorded?	*[Yes / partly / no]*
What skills or help do we still need?	*[e.g. statistician]*

4.5 Publication plan

Plan the paper now, before any data exist. Choose a journal whose scope and readers fit your question (check it is indexed and not predatory, and its open-access fee), the reporting guideline you will follow from day one, who the authors are, and what your abstract will say.

Target journal and reporting guideline

Target journal: *[First choice]* (back-up: *[second choice]*)

Why this journal: *[Scope and readers; indexed (e.g. PubMed); open access / fee; not predatory]*

Article type: *[e.g. original research, research letter, short report]*

Reporting guideline: ☐ CONSORT (trial) ☐ STROBE (observational) ☐ STARD (diagnostic) ☐ PRISMA (review) ☐ Other

Other channels: *[e.g. department meeting, conference abstract/poster, summary for staff]*

Authorship and CRediT roles

Authors must meet all four ICMJE criteria: substantial contribution; drafting or critical revision; final approval; accountability. Others go in the Acknowledgements. CRediT roles: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Author (order)	Profession	CRediT roles	Meets ICMJE 4?
1. [Name]	*Intensivist*	*Conceptualization; Methodology; Writing – original draft*	☐ *Yes*
[Name]	*[Profession]*	*[Roles]*	☐ Yes
[Name]	*[Profession]*	*[Roles]*	☐ Yes
[Name]	*[Profession]*	*[Roles]*	☐ Yes
[Name]	*[Profession]*	*[Roles]*	☐ Yes

Working title of the paper

[Refined title from 1.1: population, intervention/exposure, outcome, design]

Structured-abstract skeleton

About 250 words in the final paper. Write the Results row as what you EXPECT to report (which numbers, which effect measure), not invented results.

Heading	Our skeleton	Example (running example)
Background	*[Problem and gap in 1–2 sentences; the aim]*	*Whether restricting fluids reduces mortality in septic shock is unknown. We aimed to ...*
Methods	*[Design, setting, participants, intervention/exposure, primary outcome, analysis]*	*Pilot RCT in one ICU; adults with septic shock randomised to restrictive or usual fluids; primary outcome 28-day mortality; risk ratio with 95% CI*
Expected results	*[Which numbers you will report]*	*Number randomised and analysed; mortality n (%) per group; RR (95% CI); fluid balance median (IQR)*
Conclusion	*[What you will be able to say, matched to the design]*	*Whether a restrictive strategy is feasible, and whether a full trial is justified*

Before you submit in Session 8

- All four parts completed; no [placeholders] or grey guidance left.
- The question, objectives, primary outcome, sample size and analysis plan all match each other.
- About 6–8 pages; references in Vancouver style.
- Publication plan completed: target journal, reporting guideline, authors and CRediT roles, working title and abstract skeleton.
- 6 slides prepared from this proposal and publication plan (everyone submits them; volunteers present them in 6 minutes).